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1. Show that D_8 is not cyclic
 2. Let $\langle a^3 \rangle, \langle a^5 \rangle \leq \langle a \rangle$ where a has order 30; label these subgroups A, B .
 - (a) What is the order of A and B
 - (b) What is $A \cap B$
 - (c) if $A \cup B$ a subgroup
 - (d) what is the smallest subgroup $C \leq \langle a \rangle$ containing both A and B
 3. List all (if any) elements of order 4 in $U(20)$.
 4. If A, B are cyclic, is $A \times B$ cyclic?
 5. If A is cyclic is every subgroup cyclic?
 6. If A is not cyclic, is it possible that there is a cyclic subgroup?
 7. What is the fundamental theorem of finite cyclic groups
 8. What are two ways of defining the center of a group?
 9. Let G be a finite group and $a \in G$ such that $a^n = e$, $n \in \mathbb{N}$. Can you find a negative integer $m \in \mathbb{Z}_{<0}$ such that $a^m = e$.
 10. Give an example of groups where $Z(G) = \{e\}$, $Z(G) = G$, and $\{e\} \neq Z(G) \neq G$.
 11. What is the order of the permutation $(1\ 4\ 6\ 6\ 2)$
 12. What is the order of the permutation $(1\ 4\ 6\ 5\ 2)(9\ 8)$
 13. What is the order of the permutation $(1\ 4\ 6\ 5\ 2)(2\ 3)$
 14. If $\alpha_1, \dots, \alpha_m \in S_n$, what is

$$|\alpha_1 \cdots \alpha_m| = ?$$
 15. If $\sigma = (a_1\ a_2\ \dots\ a_n)$ is a cycle, what is σ^{-1} in cycle-representation?
 16. Show that any cycle can be broken down into a product of 2-cycles in two different ways
 17. Let S_n be the permutation group and $n \geq 3$. Show that $Z(S_n) = \{e\}$
 18. What group that you know is S_1 and S_2 isomorphic to?
 19. Show that $\exp : (\mathbb{R}, +) \rightarrow (\mathbb{R}^*, *)$ where $\exp(a) = e^a$ is a group homomorphism (hint: you can use the properties of the exponential)
 20. is $U(20)$ and $U(24)$ isomorphic?